ArcheoZS

Evaluating Geographic Generalization in Archaeological Site Detection: A Comparative Study of Foundation Models, Satellite Imagery and Bare-Earth Elevation

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Abstract

Archaeological sites are the physical record of past settlement and burial, and across the Middle East and North Africa thousands of them are being lost to looting, conflict and development before they are even mapped. Two site types dominate the region: settlement mounds (tells), built up over centuries of occupation, and burial mounds (tumuli and cairns), raised over graves. Both change the ground slightly, in its shape, its soil and its vegetation, and satellites can in principle see this over areas no field team can walk. Machine-learning studies of archaeological site detection often report high accuracy, but usually test on ground near the training data, where a model can recognise the landscape rather than the site. We test detection the hard way: 196 recorded sites (100 settlements, 96 burial mounds; two early preliminary experiments use all 197 and are labelled) from the EAMENA database across 14 regions of the Middle East and North Africa, each compared with background patches cut from the same satellite image, with whole regions at least 300 km apart held out from training. Every main result is averaged over five separate draws of backgrounds (a test of sensitivity to the background sample, not five replications of the whole pipeline) and compared with paired bootstrap tests corrected for multiple comparisons.

Four findings hold. (1) The test setup matters: the same elevation model scores an overall AUC of 0.78 when regions are mixed between training and test, but 0.66 when whole regions are held out (within-site AUC: 0.77 against 0.68). (2) Single runs are unreliable: changing only the random backgrounds moves a model's score by up to 0.07. (3) A 300-million-parameter geospatial foundation model (Prithvi-EO-2.0), frozen or fine-tuned with LoRA, does not beat a random forest on simple spectral statistics (within-site AUC 0.58–0.62 against 0.60–0.63). Four seasons of imagery help both by about 0.03–0.04, consistently but not significantly after correction. (4) Free bare-earth elevation (FABDEM) is what works. Adding it lifts the within-site AUC from 0.60 to **0.69** [0.65–0.73] (+0.09, Holm P = 0.001; unchanged when whole regions are resampled), for settlements (0.71) and burial mounds (0.68) alike. The signal sits at the 80–160 m scale of the mounds themselves, and is unchanged when buildings and trees are removed from the elevation model. Used for survey planning with a conformal guarantee of keeping 90% of sites, the best model lets surveyors skip 35% of the sampled background patches, against 10% by chance (an estimate on sampled patches, not yet on whole images).

1. Introduction

The Middle East and North Africa hold one of the densest records of archaeological heritage in the world, and much of it faces looting, conflict damage and development at a scale no field team can survey on foot (Casana, 2015). Satellite imagery covers the whole region, so automated detection is attractive, and recent studies report accuracy, F1 or detection rates from about 75% to over 90% (Tables 1 and 10).

Remote sensing of archaeology rests on a simple idea: a site leaves a trace that differs from its surroundings, and a sensor can pick up that difference. Two kinds of site matter in this paper. **Settlements**, especially *tells*, are mounds formed by the slow build-up of mud-brick, debris and rebuilding on the same

spot; they can rise metres above the plain and their soil often differs in colour and moisture from the land around. **Burial mounds** (tumuli, cairns and the grave-mound fields of places such as Bahrain) are smaller, deliberately raised heaps of earth or stone, often in large clusters. The traces fall into three groups: *relief* (the mound itself), *soil and surface colour* (disturbed or enriched ground), and *vegetation or moisture differences* that change through the seasons. Classic work mapped tells over large areas using mainly elevation and multi-date imagery (Menze & Ur, 2012). What is less clear is which of these traces a model can use reliably in a new region, which is the question this paper tests.

Those numbers are hard to interpret. In most studies the test sites lie close to the training sites, or share images with them. A model can then score well by learning what a region's landscape looks like (soil, vegetation, terrain) instead of what an archaeological site looks like. The same problem is well documented in ecology and other fields: random cross-validation on spatial data overstates performance, sometimes dramatically (Roberts et al., 2017; Ploton et al., 2020; Kapoor & Narayanan, 2023).

This paper began as ArcheoZS, a zero-shot pipeline built around a geospatial foundation model. An earlier 18-site version of it reported 100% accuracy on a three-site test set; with proper cross-validation that fell to the level of always guessing the most common class. We therefore rebuilt the evaluation first and then asked a simple question: under a test that a landscape cannot pass, what actually helps to find sites? Our contributions are:

1. **A strict and stable evaluation.** Sites are compared with ground from the same image, whole 300 km regions are held out, and results are pooled over five background draws with corrected paired tests. We measure how much easier setups inflate scores.
2. **An honest test of a geospatial foundation model.** Prithvi-EO-2.0, frozen or LoRA fine-tuned, and given one or four dates, against random-forest baselines on identical samples and folds.
3. **Seasonal imagery.** The first comparison, to our knowledge, of one date against four seasons for archaeological detection under region-held-out testing.
4. **Bare-earth elevation as the key input.** A +0.09 AUC gain that survives correction, is traced to mound-sized relief, and is unchanged after removing buildings and trees.
5. **Decisions with guarantees.** Yes/no decisions calibrated across regions, and conformal survey triage that states how much ground can be skipped while keeping a chosen share of sites.

2. Related Work

Machine-learning approaches to archaeological site detection have moved from statistical classification to convolutional networks for single site types, to UAV and time-series methods, and most recently to pretrained foundation models. Table 1 summarises ten representative studies.

Table 1. A decade of machine-learning approaches to archaeological site detection.

Year	Study	Method	Reported limitation	Venue
2012	Menze & Ur	Random forest on multitemporal ASTER imagery and SRTM elevation	Detects anthrosols and mounds; no site-type discrimination	PNAS 109(14)
2019	Caspari & Crespo	CNN binary classifier on satellite patches	Presence/absence only; no localisation	J. Archaeol. Sci. 110
2020	Orengo et al.	SAR + optical fusion, multitemporal, Google Earth Engine	Tuned to mound shapes in arid terrain	PNAS 117(31)

Year	Study	Method	Reported limitation	Venue
2021	Chen et al.	CNN object detection (two-stage detectors)	High-resolution imagery; Altai Mountains only (four regions plus one unsurveyed area)	ISPRS J. Photogramm. 177
2023	Barragán et al.	Image processing + classifier	Engineered for dense vegetation	Appl. Sci. 13(11)
2023	Altaweel et al.	Deep learning on UAV imagery (looting monitoring)	Needs drone resolution	Soc. Sci. Comput. Rev.
2023	Payntar	Landsat time series + random forest	Measures destruction of known sites; does not detect new sites	SAA 2023 (abstract)
2025	Vincent et al.	Foundation model + temporal attention	Needs multi-year monthly series	CVPRW 2025
2025	Kozioł et al.	ML-predicted survey visibility	Schedules surveys; does not detect sites	Remote Sens. 17(13)
2026	Chiricallo et al.	YOLOv8-Seg on Sentinel-1 radar + Copernicus 30 m elevation	Tells only, one region; needs fine-tuning in Iran	Remote Sens. 18(13)

Entries checked against publisher, repository or conference records, 2012–2026. Chen et al., Altaweel et al. and Payntar were checked at abstract level only.

2.1 Positioning of this work

Two strands of this literature matter most here. First, Menze & Ur (2012) and Orengo et al. (2020) showed that settlement mounds can be mapped from elevation and multi-date imagery in Mesopotamia and the Indus region, but evaluated within the regions they mapped. Chiricallo et al. (2026) extend this to deep learning on free Sentinel-1 radar and the Copernicus 30 m elevation model, segmenting tells in central Iraq; in their control experiments elevation alone (average precision 0.44) far outperformed radar alone (0.12) or optical RGB alone, and a model trained in Iraq lost accuracy in Iran without regional fine-tuning. Second, recent work applies foundation models to archaeological imagery (Vincent et al., 2025). We bring the two together under one strict protocol: the same sites, backgrounds and folds for spectral baselines, a foundation model, seasonal imagery and elevation, so the contribution of each can be measured rather than assumed.

3. Data

3.1 Sites and labels

Site records come from the EAMENA database (Endangered Archaeology in the Middle East and North Africa), retrieved as CIDOC-CRM linked data. Each record's own type and form descriptions were mapped to two classes with fixed rules: *tells* and records described as settlement or habitation became **settlements**; records described as tomb, grave or burial together with a funerary form (mound, cairn, tumulus) became **burial mounds**; all other records were dropped rather than guessed. This gave 197 sites (101 settlements, 96 burial mounds). **Two site counts appear in this paper.** The first experiments (site-type matching in Section 5.1 and the first Prithvi comparison in Section 5.3) were run early, before the seasonal imagery existed, on all 197 sites. All later experiments, including every headline number and the abstract, use **196 sites (100 settlements, 96 burial mounds)**, because one settlement was dropped when its image window was not clean in every season. Whenever a section uses the 197-site set it says so.

The sites span 18 countries, most in Libya (44), Iraq (32), Bahrain (28, the Dilmun burial-mound fields), Yemen (26), Sudan (12), Jordan (10) and Syria (10). Sites closer than 300 km to each other are joined into one region (single linkage), giving 14 regions (Figure 1); the largest holds 58 sites.

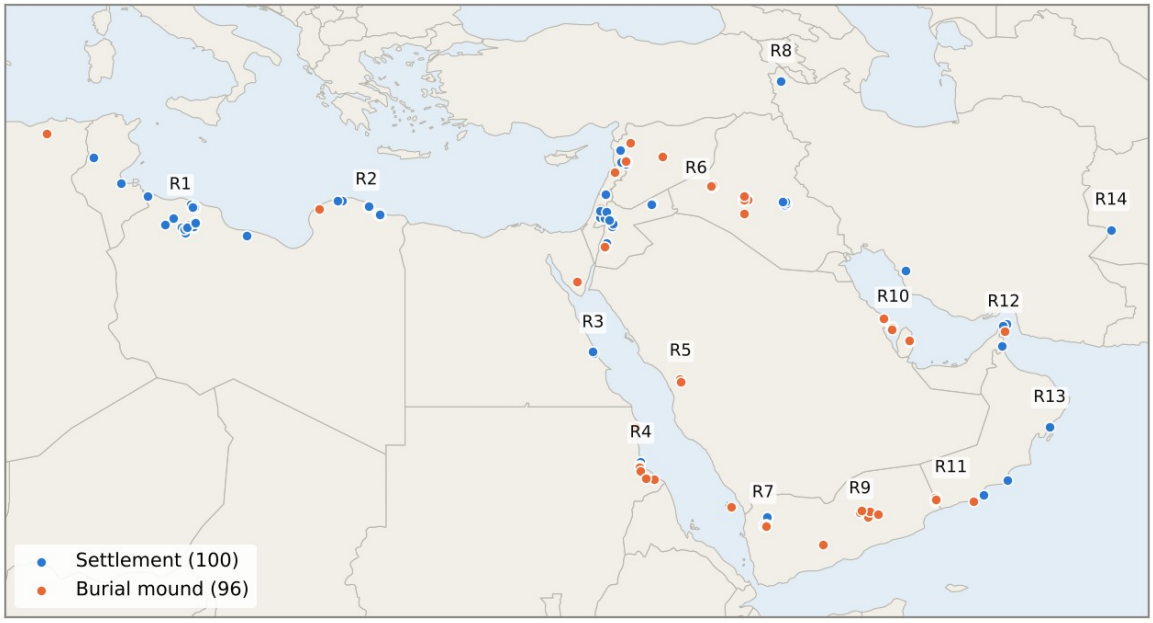


Figure 1. The 196 sites used in the main experiments, coloured by type. Labels R1–R14 mark the regions used for cross-validation: all sites of a region are always held out together. Borders: Natural Earth. Coordinates: EAMENA. Exact positions are deliberately shown at small scale.

3.2 Imagery

Sentinel-2 Level-2A imagery was retrieved from the Copernicus Data Space Ecosystem for 73 tiles covering all sites. We use the six bands Prithvi-EO-2.0 was pretrained on: B02, B03, B04, B8A, B11 and B12 (blue, green, red, narrow near-infrared, two short-wave infrared). The original composites are medians over January–February 2024. For the seasonal experiments, the same areas were downloaded as median composites for April–June, July–September and October–December 2024 (219 additional files).

The winter images are not clean. Using a simple brightness check (median blue reflectance above 2,500), 9 of the 73 January–February tiles appear partly covered by cloud or snow, against 1, 2 and 0 tiles in the other three seasons. A single-date model in this region therefore sees about one image in eight partly obscured.

3.3 Elevation

Two elevation models were resampled onto each tile's own 10 m grid (bilinear): the Copernicus GLO-30 model, which measures the **surface** including buildings and trees, and FABDEM (Hawker et al., 2022), the same model with buildings and trees removed by machine learning, i.e. a **bare-earth** model. Both have a native resolution of about 30 m. The 10 m grid only aligns elevation with the imagery and adds no detail: the effective resolution of every elevation feature stays about 30 m (roughly three image pixels), so even the 80 m window covers only a few elevation cells. Unless stated otherwise, results use FABDEM.

3.4 Backgrounds and reference patches

For every site, three background patches are cut from the same image, centred at least 2.5 km from any known site. A location (site or background) is used only if its full 224 × 224 window (2.24 km) lies inside the image in all four seasons with at most 5% empty pixels. The same rule applies to sites and backgrounds, so missing data cannot become a shortcut. Because the backgrounds are random, results depend on which

ones are drawn, so we built five separately drawn sets (random seeds 123, 7, 11, 19 and 42), giving 15 backgrounds per site in total. These five draws test how sensitive the results are to the choice of backgrounds. They are not five independent replications of the whole pipeline: the sites, regions, folds, features and models are identical in every draw, and only the backgrounds change. A fingerprint of each patch is stored with its elevation data, so every experiment provably uses identical samples. Separately, 25 random, unlabeled reference patches per image are used only to calibrate yes/no decisions (Section 4.4). Figure 2 shows two sites with their backgrounds.

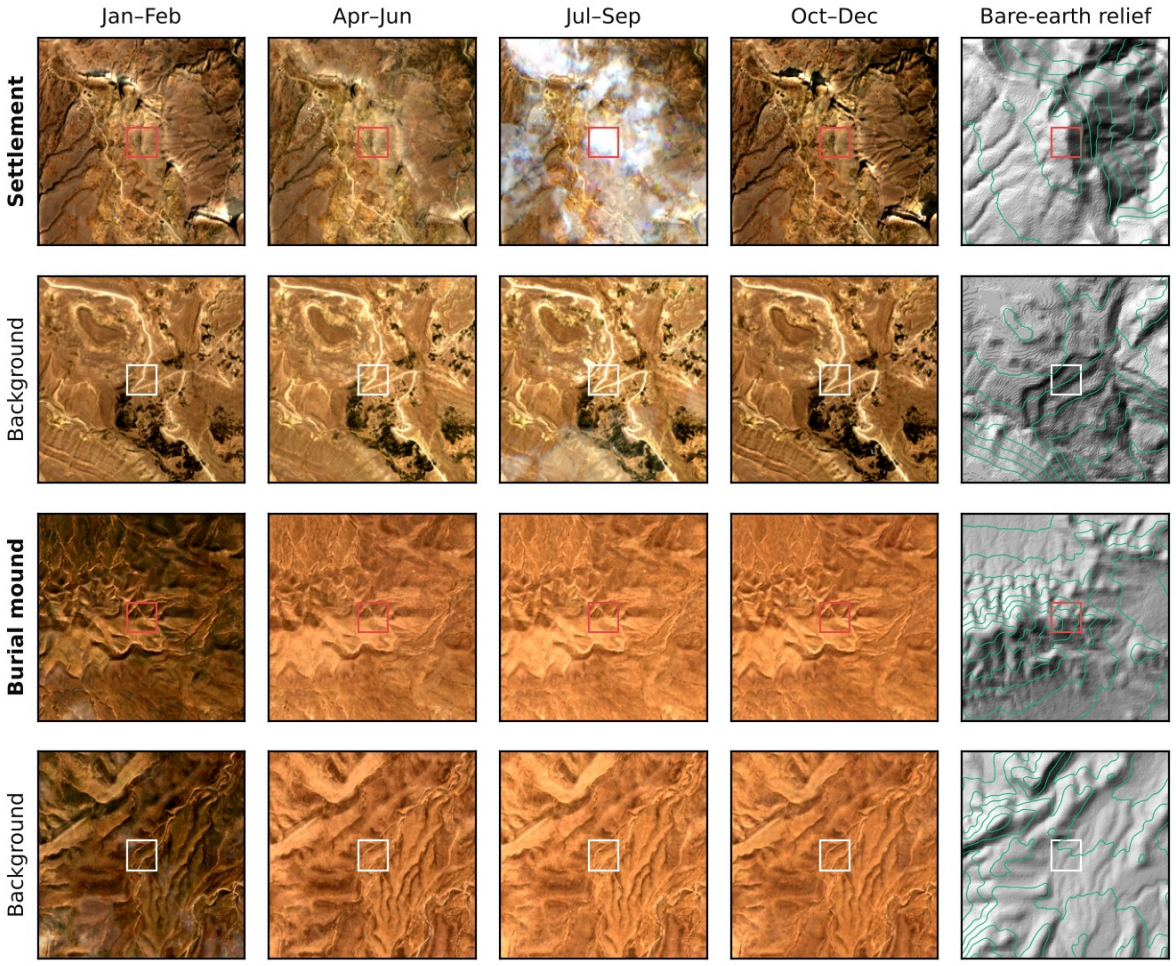


Figure 2. A settlement (EAMENA-0095558) and a burial mound (EAMENA-0271076), each with one background patch from the same image, in the four 2024 seasons (natural colour) and as bare-earth relief (FABDEM hillshade with contours). Note the cloud in the July–September settlement image and how much the surface colour changes through the year.

4. Methods

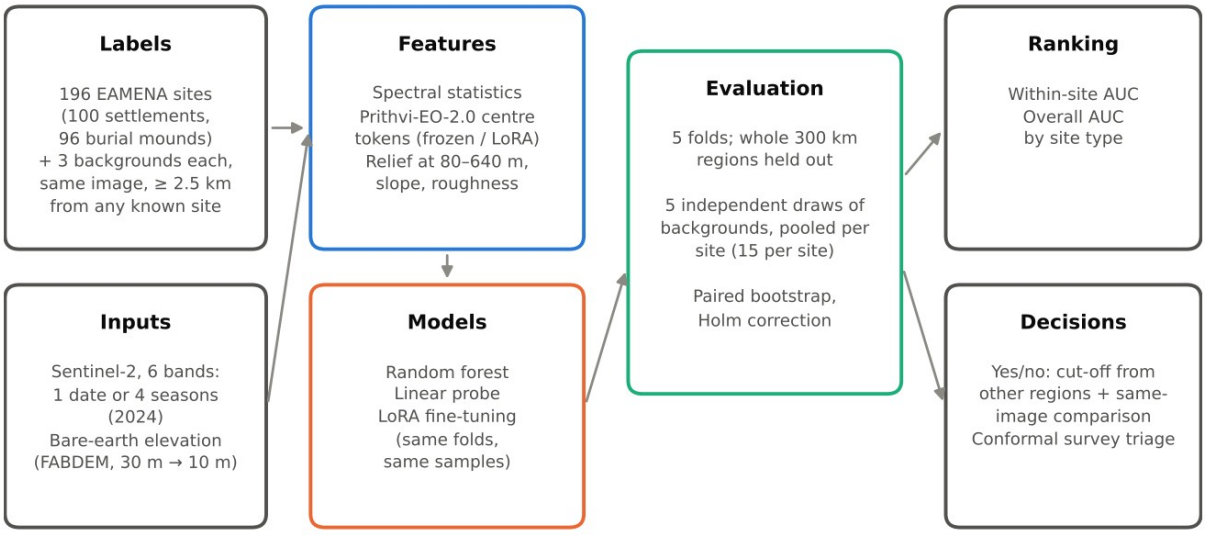


Figure 3. Study design. All models see identical samples and folds; every comparison is paired by site.

4.1 Features

- **Spectral statistics.** For each band, the mean, standard deviation and 10th and 90th percentiles of the central 320 m and of the whole patch, plus the centre-minus-patch contrast (54 values per date). The four-season version concatenates all four dates and adds, for each band, how much the centre varies over the year (228 values).
- **Prithvi-EO-2.0-300M** (Szwarcman et al., 2024), a Vision Transformer pretrained on Harmonized Landsat–Sentinel time series. Patches are standardised with the model's own pretraining statistics. We use the mean of the central 2 × 2 tokens (about 320 m), because the summary (CLS) token was almost identical for all sites (cosine similarity 0.995). For four seasons the model reads all dates at once, as in its pretraining, giving one feature vector per season; we report both their mean and their concatenation.
- **Elevation** (15 values, all relative, never absolute altitude, so a region's height cannot be learned as a shortcut): how far the centre is raised above the ring around it, and how far its highest point rises above the ring's median, for central windows of 80, 160, 320 and 640 m; slope and roughness in the centre and the whole patch; the centre's height relative to the whole patch; and the share of the patch lower than the centre.

4.2 Models

Random forest: 500 trees, minimum leaf size 2, balanced class weights. **Linear probe:** logistic regression (C = 0.1) on standardised features, balanced class weights. **LoRA fine-tuning:** rank-8 adapters on Prithvi's attention projections (786,432 trainable parameters, 0.26% of the model) with a linear head, trained with a weighted binary cross-entropy loss (Adam, learning rate 10^{-4}, 8 epochs). A second LoRA run added random rotations and flips, weight decay (0.01) and early stopping on 15% of the training sites (up to 15 epochs). Hyperparameters were set once from common defaults (random forest: 500 trees, minimum leaf size 2; linear probe: C = 0.1) and were not tuned: no search was run on any data, so no information from

the test regions could enter them. Tuning inside the training regions might change the scores, in either direction.

4.3 Evaluation

Five folds are formed from whole regions (grouped cross-validation), so no test site has a training site within 300 km. Two scores are reported, both with chance at 0.50. **Within-site AUC** is the share of (site, own background) pairs in which the site scores higher; since both come from the same image, regional differences cancel out. **Overall AUC** ranks all sites against all backgrounds. For the main results, each site's within-site AUC is averaged over the five background draws; 95% intervals come from 5,000 bootstrap resamples of sites, and methods are compared with paired bootstrap tests on the same sites. The comparisons were fixed before the pooled results were seen, and each family of tests is corrected with the Holm method; we call a difference convincing only if its corrected one-sided P is below 0.025. Because regions differ greatly in size (from 1 to 58 sites), fold composition, per-fold scores and a region-level bootstrap are reported in Appendix A. A second analysis that resamples whole regions instead of sites is in Appendix B.

4.4 Yes/no decisions

A ranking is not a decision. To turn scores into site/not-site calls in an unseen region, the cut-off for each fold is chosen on the predictions of the other folds only (maximising balanced accuracy). Because scores also drift between images, we additionally express each score relative to the scores of the 25 unlabeled reference patches from the same image (a z-score). This uses no labels from the test region: in real use, random ground from the same image is always available.

4.5 Conformal survey triage

For survey planning we ask: if a team only visits ground the model flags, what share of sites will they still find, and how much ground can they skip? For each held-out fold, a threshold is set at the conformal lower quantile of the site scores in the other folds (Vovk et al., 2005; Angelopoulos & Bates, 2021), which guarantees, under exchangeability, that at least the target share of sites score above it. We report the share of held-out sites actually kept (to check the guarantee across regions) and the share of sampled background patches below the threshold (the ground that, on this sample, can be skipped). A model with no skill can skip only one minus the target.

5. Results

5.1 Easier test setups overstate performance

Table 2 and Figure 4 evaluate the same elevation model three ways. Splitting at random lets a site and its own backgrounds be separated, which mixes the two scores in an unusual way; keeping each site with its backgrounds but mixing regions is the setup most studies use. Against that common setup, holding out whole regions lowers the overall AUC from 0.78 to 0.66 and the within-site AUC from 0.77 to 0.68. Some of what an easier test credits to the model is knowledge of the region.

Table 2. One model (random forest on bare-earth elevation), three test setups. Mean over five background draws (range).

Test setup	Within-site AUC	Overall AUC
A. Random patches (a site and its own background can be split)	0.722 (0.71–0.73)	0.738 (0.73–0.75)

Test setup	Within-site AUC	Overall AUC
B. Sites kept with their backgrounds, regions mixed	0.774 (0.76–0.80)	0.777 (0.77–0.79)
C. Whole 300 km regions held out (this paper)	0.682 (0.66–0.71)	0.661 (0.64–0.68)

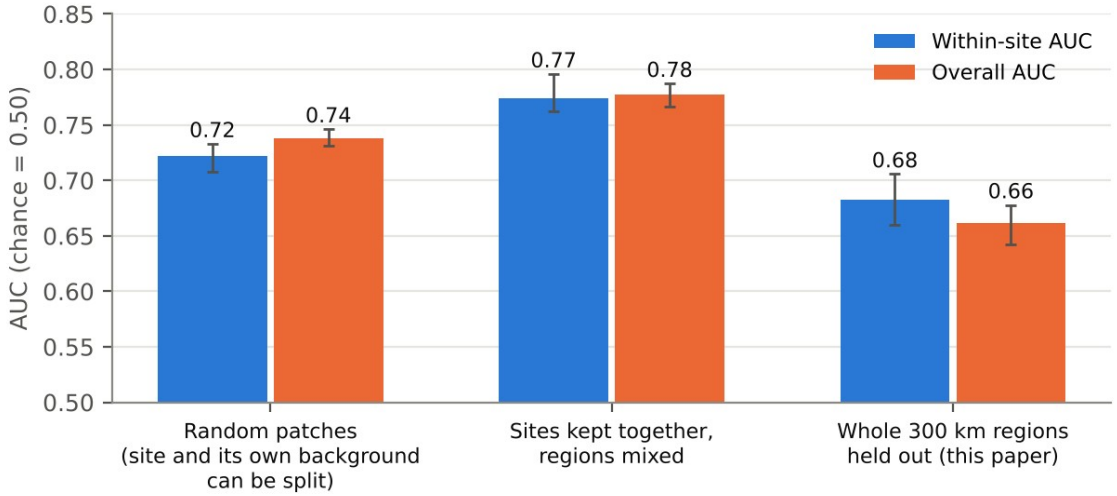


Figure 4. The same model scored under three test setups. Bars show the mean of five background draws; whiskers show their range.

The same effect appeared when classifying site type. Comparing each site with its most similar site in Prithvi's feature space, 116–121 of the 197 sites (59–61%; the original set, before one site was excluded) were matched to a site of the same type when only sites within 50 km were held out together, above the 51% obtained by always guessing the commoner type. With 300 km regions held out, this fell to 80–92 of 197 (41–47%), below that baseline, and matching each site to the nearest class average fell from 59–103 to 39–48 of 197. Site type (settlement or burial mound) could not be predicted from these images at all, so the rest of the paper addresses detection: telling a site from ordinary ground.

5.2 Single runs are unreliable

Figure 5 shows each method's within-site AUC in each of the five runs that differ only in which backgrounds were drawn. Scores move by 0.03–0.07, and Prithvi with four seasons moves from 0.58 to 0.65. Two runs of the same experiment can therefore point in opposite directions: in our first background draw, four seasons helped the random forest significantly but not Prithvi; in the second, the reverse. This spread is larger than many improvements reported in the literature from a single run. All main results below are therefore pooled per site over the five draws.

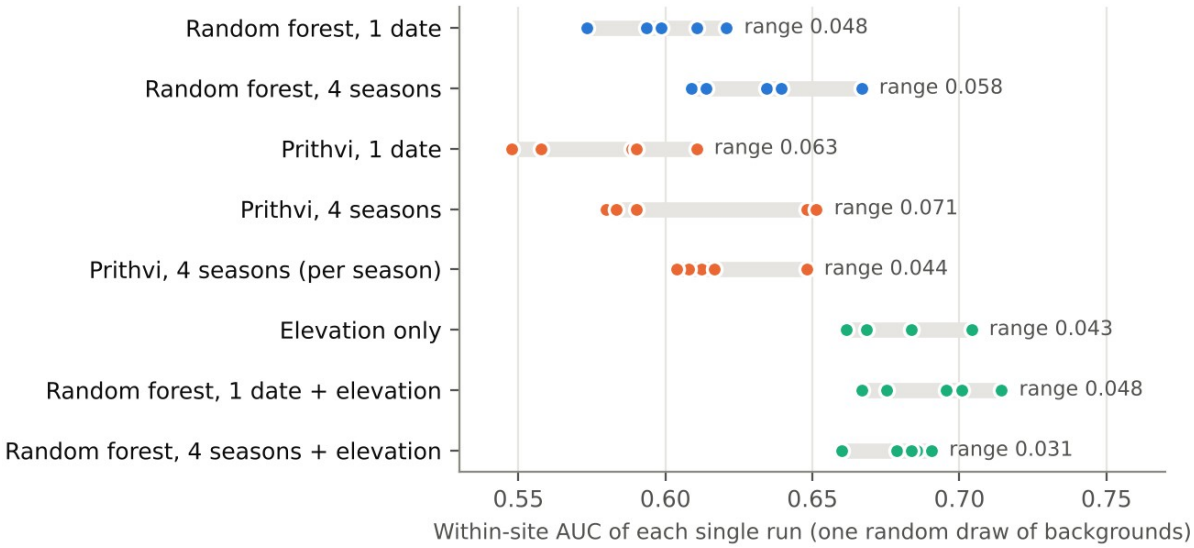


Figure 5. Within-site AUC of every single run. Each dot is one of five separate draws of backgrounds; the grey band spans their range.

5.3 Preliminary experiment (197 sites, one background draw): a geospatial foundation model does not beat a simple baseline

This is a preliminary experiment, run before the seasonal imagery existed. It is kept separate from the main 196-site results in Section 5.4, which supersede it. Table 3 compares Prithvi-EO-2.0 with the spectral random forest on one date, on the original 197-site set with one draw of backgrounds (the 196-site, five-draw results follow in Section 5.4). Frozen Prithvi and LoRA fine-tuning score slightly above the random forest on within-site AUC, but no difference is significant. LoRA added nothing over the frozen model, and its overall AUC was the lowest (0.55–0.58), meaning its scores do not transfer between regions. In the second LoRA run the training loss kept falling while validation scores did not, a sign of overfitting with about 160 training sites.

Table 3. Foundation model and fine-tuning, one date, 197 sites, one background draw, 300 km regions held out.

Method	Within-site AUC [95% CI]	Overall AUC [95% CI]
Random forest, spectral (baseline)	0.591 [0.54–0.64]	0.596 [0.55–0.64]
Prithvi, frozen + linear probe	0.618 [0.57–0.66]	0.612 [0.57–0.65]
Prithvi + LoRA (8 epochs)	0.624 [0.58–0.67]	0.582 [0.54–0.62]
Prithvi + LoRA (augmentation, early stopping, weight decay)	0.607 [0.55–0.66]	0.553 [0.51–0.59]

Paired differences in within-site AUC (second run): frozen Prithvi minus random forest +0.027 [−0.027 to +0.083]; LoRA minus random forest +0.017 [−0.042 to +0.078]; LoRA minus frozen −0.010 [−0.069 to +0.049]. None is significant.

Figure 6 shows how each method improves with more training sites. The random forest stays flat at about 0.58–0.59 from 39 to 158 training sites. Frozen Prithvi rises steadily from 0.52 to 0.62 and has not levelled off, so a foundation model may overtake simple features with several hundred more sites. LoRA is erratic. With the data available, though, the foundation model offers no measurable advantage.

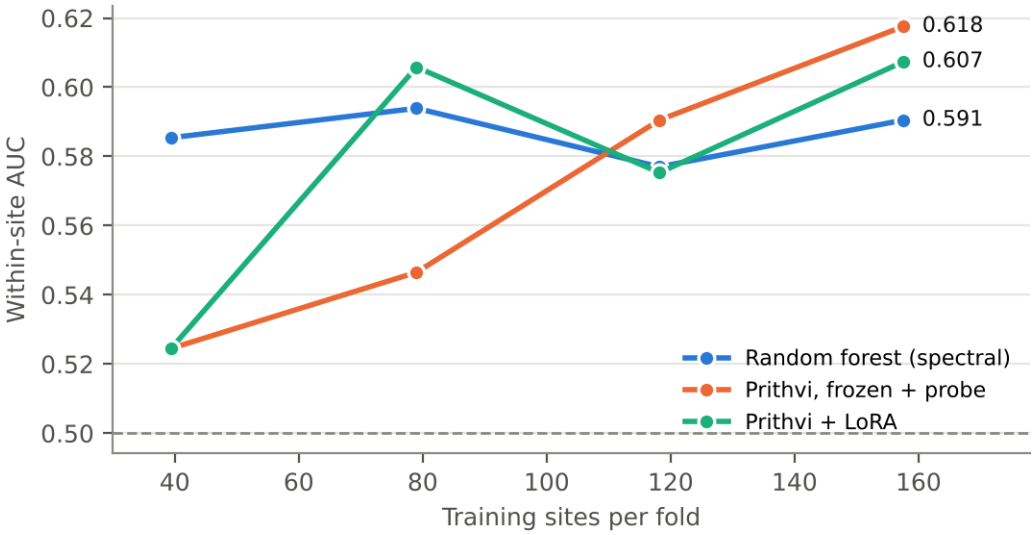


Figure 6. Within-site AUC against the number of training sites per fold (one date; nested subsets of 25–100% of the training sites; the test folds are always complete). Intervals are about ±0.05 and are omitted for clarity.

5.4 Seasons help a little; elevation helps clearly

Figure 7 and Table 4 give the main results, pooled over five background draws on the same 196 sites, regions and folds. Four seasons raise both the random forest (0.599 to 0.633) and frozen Prithvi (0.579 to 0.611). Adding bare-earth elevation gives the largest gain of any change we tested: from 0.599 to **0.690** [0.65–0.73] for the one-date random forest. Elevation alone reaches 0.680, more than any colour-based model.

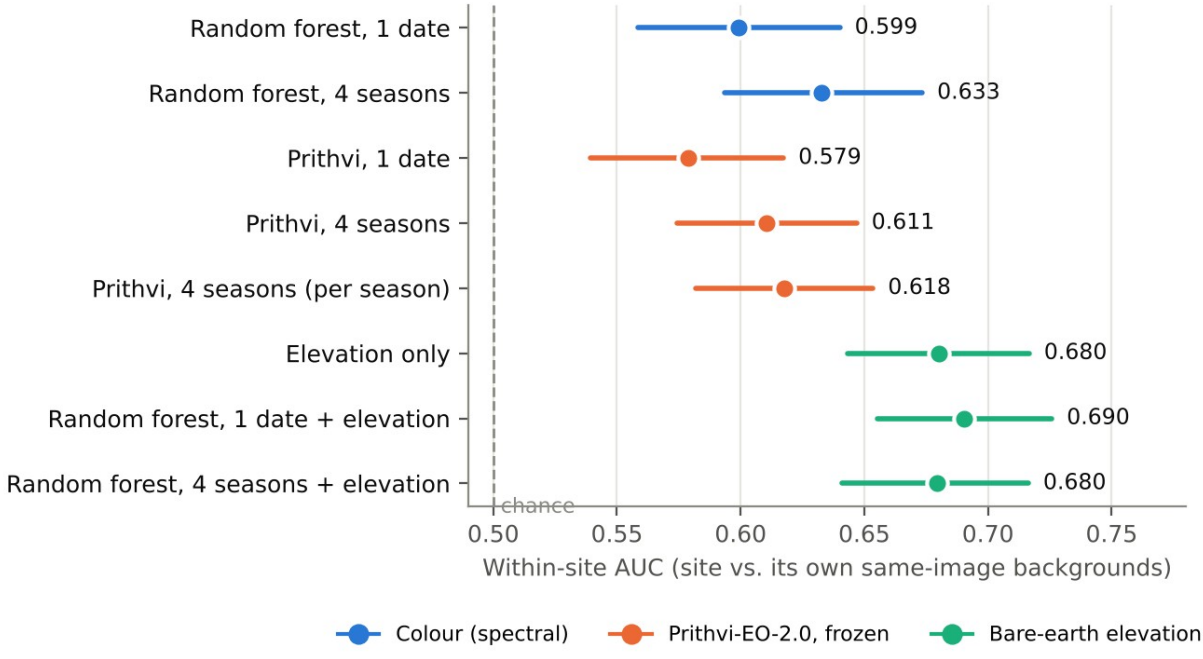


Figure 7. Main results: within-site AUC pooled per site over five background draws (196 sites, 300 km regions held out), with 95% bootstrap intervals.

Table 4. Main results, pooled over five background draws, 196 sites, 300 km regions held out.

Method	Within-site AUC [95% CI]	Settlements	Burial mounds	Overall AUC
Random forest, 1 date	0.599 [0.56–0.64]	0.619	0.579	0.607
Random forest, 4 seasons	0.633 [0.59–0.67]	0.658	0.606	0.623
Prithvi (frozen), 1 date	0.579 [0.54–0.62]	–	–	0.580
Prithvi (frozen), 4 seasons	0.611 [0.57–0.65]	–	–	0.602
Prithvi (frozen), 4 seasons, per season	0.618 [0.58–0.65]	–	–	0.601
Elevation only (FABDEM)	0.680 [0.64–0.72]	0.693	0.667	0.664
Random forest, 1 date + elevation	**0.690 [0.65–0.73]**	**0.705**	**0.675**	**0.681**
Random forest, 4 seasons + elevation	0.680 [0.64–0.72]	0.693	0.666	0.662

Overall AUC is the mean over the five draws. Random-forest rows come from the elevation experiment, which recomputes the spectral features; the seasonal experiment gave 0.596 and 0.633 for the same two rows. Results by site type were not computed for Prithvi.

Table 5. Paired comparisons on the same sites (within-site AUC, pooled over five draws). Holm correction within each family.

Comparison	Difference [95% CI]	P	Holm P	Convincing?
Prithvi: 4 seasons minus 1 date	+0.032 [−0.006 to +0.069]	0.051	0.152	No
Random forest: 4 seasons minus 1 date	+0.037 [+0.003 to +0.071]	0.017	0.067	No
4 seasons: Prithvi minus random forest	−0.022 [−0.067 to +0.023]	0.829	1.000	No
4 seasons, per season: Prithvi minus random forest	−0.015 [−0.057 to +0.028]	0.749	1.000	No
1 date: + elevation minus without	**+0.091 [+0.059 to +0.125]**	<0.001	0.001	**Yes**
4 seasons: + elevation minus without	**+0.047 [+0.024 to +0.069]**	<0.001	0.001	**Yes**
Elevation only minus chance (0.50)	**+0.180 [+0.142 to +0.216]**	<0.001	0.001	**Yes**

The seasonal gain is consistent (four seasons beat one date for the random forest in all five draws) but does not survive correction, so we describe it as a likely but unconfirmed improvement; resampling whole regions (Appendix B) leaves it unconfirmed under the more cautious of two region-level tests. The elevation gains are large and convincing. Combining all inputs did not beat one date plus elevation: the 228 seasonal features dilute the 15 elevation features in the random forest, so better ways of combining them remain to be tested.

5.5 What elevation detects

Two checks suggest that the elevation signal comes from the archaeological mounds themselves. **First, scale (Figure 8).** Whether the centre is raised above its surroundings separates settlements from their backgrounds best in an 80 m window (AUC 0.67), less at 160 m (0.63) and 320 m (0.55), and not at all at 640 m (0.50). The signal is local, at the size of a tell, not a preference for high ground. For burial mounds, the height of the highest point above the surroundings separates best (0.70–0.73), as expected for small, steep tumuli. In bare-earth elevation, sites rise a median of 0.25 m above the ring 160 m around them, while backgrounds rise 0.00 m.

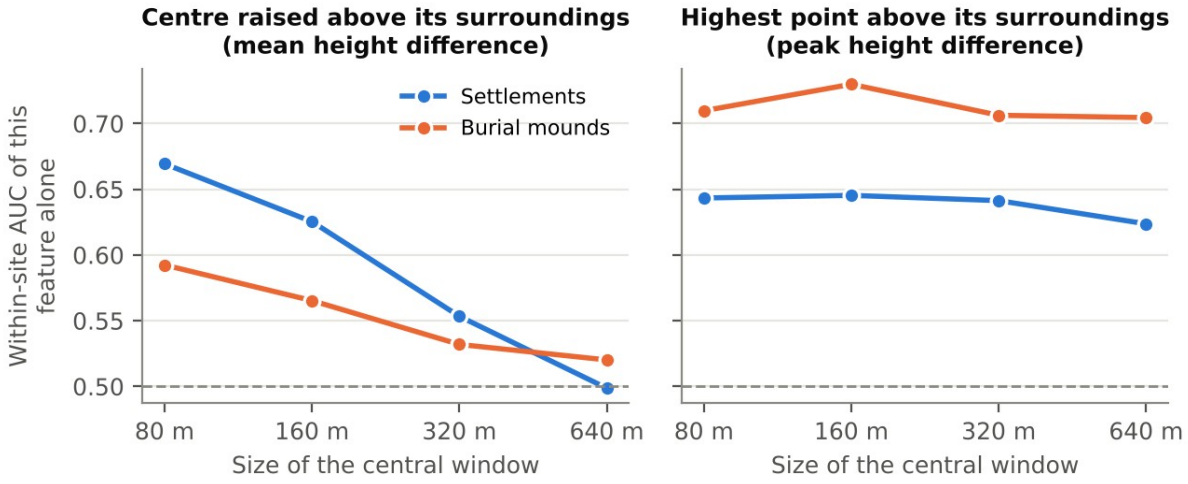


Figure 8. How well each elevation feature alone separates sites from their own backgrounds (within-site AUC, pooled over five draws), by size of the central window and site type.

Second, buildings and trees (Table 6). Modern villages often stand on or beside ancient tells, so a surface elevation model could detect modern buildings. Replacing it with the bare-earth model changed every result by at most 0.01, with every interval spanning zero. The signal is in the ground itself, not in modern buildings. What these checks do not exclude is that natural relief (dunes, wadi banks, rock outcrops) produces a similar signal: our backgrounds are random ground, not natural mounds or terrain-matched ground, so the model may partly separate any raised or rough ground from flat ground. We have not yet tested this (Section 8).

Table 6. Surface elevation (Copernicus GLO-30) against bare-earth elevation (FABDEM), same sites and backgrounds, pooled over five draws.

Method	Bare earth (FABDEM)	Surface (Copernicus)	Difference [95% CI]
Elevation only	0.680	0.690	−0.010 [−0.037 to +0.016]
Random forest, 1 date + elevation	0.690	0.681	+0.009 [−0.010 to +0.029]
Random forest, 4 seasons + elevation	0.680	0.675	+0.005 [−0.012 to +0.022]

5.6 From ranking to decisions

Yes/no decisions (Table 7, Figure 9). With the usual cut-off of 0.5, every model is barely better than chance (balanced accuracy 0.52–0.55), because its scores shift between regions. Choosing the cut-off on other regions raises this to 0.55–0.63, and comparing each score with random ground from the same image adds a further 0.01–0.02. The best model reaches a balanced accuracy of **0.65**.

Table 7. Balanced accuracy of yes/no decisions in unseen regions (chance 0.50), mean of five draws.

Method	Cut-off 0.5	Cut-off from other regions	+ same-image comparison
Random forest, 1 date	0.519	0.550	0.561
Random forest, 4 seasons	0.530	0.580	0.598
Elevation only	0.545	0.609	0.621
Random forest, 1 date + elevation	0.520	0.625	**0.647**

Method	Cut-off 0.5	Cut-off from other regions	+ same-image comparison
Random forest, 4 seasons + elevation	0.534	0.609	0.631

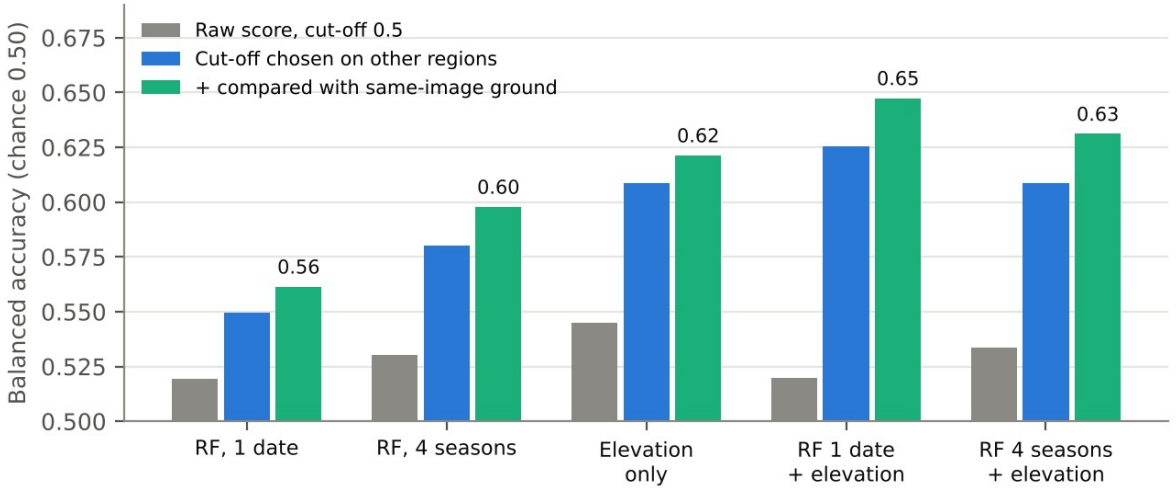


Figure 9. Balanced accuracy of yes/no decisions in unseen regions under three ways of setting the cut-off.

Survey triage (Table 8, Figure 10). The conformal guarantee held across regions: with a 90% target, the share of held-out sites kept was 89–91% for every method. At that target the colour-only models let surveyors skip about 21–22% of the sampled background patches, against 10% by chance. With elevation this rises to 30%, and to **35%** when it is combined with one date of imagery (32–41% across draws).

These shares refer to the sampled background patches (random 2.24 km patches from the same images, at least 2.5 km from any recorded site). They are not shares of a whole area, because the backgrounds may contain unrecorded sites; they become area estimates only if the model is run over whole images. The guarantee also assumes exchangeability: the scores of held-out sites are treated as if drawn from the same distribution as those of the calibration sites. Regions differ strongly (Appendix A), so this is an assumption, not a fact. The kept share (89–91%) is the empirical coverage over all held-out sites together; we did not test it within each region, where it is noisy (with 34 sites and a 90% target, chance variation alone has a standard deviation of about 5 percentage points).

Table 8. Conformal survey triage: share of held-out sites kept and of sampled background patches that can be skipped, mean of five draws (range of ground skipped).

Method	Target 90%: sites kept	Backgrounds skipped	Target 80%: sites kept	Backgrounds skipped
No skill (chance)	90%	10%	80%	20%
Random forest, 1 date	90.0%	21.3% (18–23%)	80.9%	31.7%
Random forest, 4 seasons	90.7%	21.7% (19–26%)	79.7%	36.0%
Elevation only	89.1%	30.3% (28–32%)	80.5%	41.0%
Random forest, 1 date + elevation	90.4%	**35.2% (32–41%)**	79.8%	**45.6%**

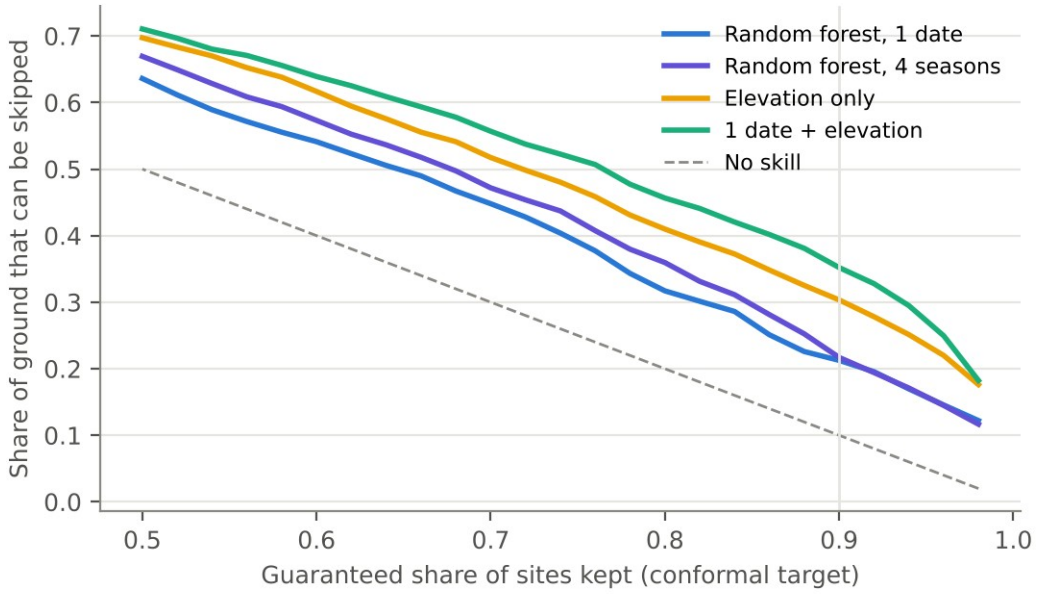


Figure 10. Share of sampled background patches that can be skipped (y) while keeping a guaranteed share of sites (x), mean over five draws. The dashed line is a model with no skill; the vertical line marks the 90% target.

5.7 Faults found and corrected

Several faults produced misleading numbers during development. We list them because each is easy to repeat in small remote-sensing studies; results reported before their correction should be disregarded.

- **Wrong input bands.** Prithvi's band list (B02–B07) follows Landsat-style naming; requesting Sentinel-2 bands B02–B07 returned three red-edge bands instead of the infrared ones, with no error raised. The correct bands are B02, B03, B04, B8A, B11 and B12.
- **Missing normalisation.** Patches were fed as raw reflectance; all stages now use the pretraining means and standard deviations.
- **Small test splits and leakage.** A three-site test split gave 100% accuracy; overlapping patches were split across folds; an unseeded split silently overrode the seeded one.
- **Landscape shortcuts.** Classifying site type looked above baseline with 50 km grouping and fell below it with 300 km grouping (Section 5.1).
- **Truncated downloads.** Two image tiles were cut off mid-download, silently removing 35 burial mounds; every tile is now read in full before use.
- **Single-run conclusions.** One background draw made seasons look significant for one method and not the other; the next draw reversed it. Pooling over five draws resolved this (Section 5.2).

6. Discussion

6.1 What the results show

Under a test that the landscape cannot pass, archaeological sites at 10 m resolution are detectable but hard to detect. Colour alone gives a within-site AUC of about 0.60, and neither a 300-million-parameter foundation model nor fine-tuning it changes that with about 200 sites. What changes it is physical: settlement mounds and burial tumuli are raised above the ground around them, and free bare-earth

elevation captures this even though its native resolution (30 m) is coarser than the imagery. This agrees with Menze & Ur (2012), who mapped Mesopotamian settlement mounds largely from elevation, and with Chiricallo et al. (2026), whose deep-learning tell segmentation in Iraq found elevation to be the primary signal, ahead of Sentinel-1 radar and Sentinel-2 colour; we show that the idea transfers to unseen regions across the Middle East and North Africa, and for burial mounds as well as settlements.

The foundation-model result should be read with its learning curve. Prithvi was pretrained on 30 m imagery and on land-cover-type tasks; archaeological sites are small and rare targets. Its frozen features still improve with more sites while the spectral baseline does not, so its advantage may appear with a few hundred more sites. The present data do not show it, and we make no claim that they do.

6.2 Practical use

A ranking with AUC 0.69 is not a site finder, but it is useful for planning. A survey team covering only the ground our best model flags would, with a guarantee that held in every held-out region, still find about 90% of known site types while skipping about a third of the sampled background patches (not yet tested on whole images, where unrecorded sites and varied terrain would change the figure). The calibrated yes/no decisions, at 0.65 balanced accuracy, are better suited to prioritising than to replacing field checks.

6.3 Positioning relative to prior work

Table 9 places these results beside recent studies. The rows are not directly comparable: tasks, data sizes, resolutions and, above all, test setups differ. Our numbers are lower than many reported elsewhere; Section 5.1 shows that a test setup like the common one would raise ours by about 0.1 AUC.

Table 9. This work beside recent related studies.

Study	Model	Data	Test setup	Reported performance
This work	Random forest, 1 date + bare-earth elevation	EAMENA, 196 sites, 14 regions, 18 countries	300 km regions held out; same-image backgrounds; 5 draws	Within-site AUC 0.69; balanced accuracy 0.65; 35% of sampled background patches skipped at 90% of sites kept
This work	Prithvi-EO-2.0 (frozen / LoRA)	Same	Same	Within-site AUC 0.58–0.62, not above the spectral baseline
Vincent et al. (2025)	DOFA foundation model + temporal attention	675 Afghan sites (looting)	Held-out test set, n = 61	F1 74.9%, AUROC 87.1%
Orengo et al. (2020)	Random forest, SAR + optical, multitemporal	Cholistan mounds	5 training and 20 validation mounds (large, well preserved); legacy survey comparison	No accuracy metric reported; probability map over about 36,000 km²
Menze & Ur (2012)	Random forest, ASTER + SRTM	Northern Mesopotamia	6 × 6 km spatial blocks within each image scene	About 90–94% of field-surveyed sites flagged; about 14,000 candidate mounds mapped
Chiricallo et al. (2026)	YOLOv8-Seg, Sentinel-1 + COP-30 elevation	Tells, central Iraq (290 test sites); Iran transfer	Separate test areas within one region	AP 0.495, IoU 0.36 (elevation alone AP 0.44)

Performance figures for prior work are as reported by the original authors. Prior-work entries were checked against the paper, abstract or repository record named in the reference list.

6.4 Comparison under the same metrics

Table 9 lists each study's own metric. To compare like with like, we recomputed our results under the metrics that other studies report (Table 10). Our yes/no decisions use a cut-off chosen on the other folds, so test regions never influence it. Precision, F1 and accuracy depend on how common the positive class is, so we give them at our natural mix of one site to three backgrounds and at a one-to-one mix, which is close to the class balance in Vincent et al. (26 of 61 test sites). AUROC and balanced accuracy do not depend on the mix. All values are means over the five background draws.

Table 10. Our headline model (random forest, 1 date + bare-earth elevation) under the metrics reported by other studies.

Study	Test setup	Their result	Ours, same metric
Vincent et al. (2025)	By site; spatially limited test set, n = 61	Accuracy 0.79; F1 0.75; AUROC 0.87; false-alarm rate 0.19	Accuracy 0.62; F1 0.67; AUROC 0.68; false-alarm rate 0.52 (1:1 mix)
Caspari & Crespo (2019)	Random 75/25 split of 1,212 images	F1 0.91 (CNN, images with tombs)	F1 0.67 (1:1 mix); 0.46 (1:3 mix)
Barragán et al. (2023)	Random 5-fold; one labelled image	Accuracy 0.94–0.97; F1 0.94–0.95	Accuracy 0.62 (1:1 mix); F1 0.67
Menze & Ur (2012)	6 × 6 km spatial blocks	Detection rate about 90–94% of surveyed sites	Recall 0.76 at a 0.52 false-alarm rate
Chiricallo et al. (2026)	Separate test areas, one region	Average precision 0.495; IoU 0.36	Not applicable (they segment outlines; we classify patches)

Our values: accuracy, precision, recall, F1 and false-alarm rate use a cut-off chosen on the other folds. At the natural 1:3 mix: accuracy 0.55, precision 0.33, recall 0.76, F1 0.46, false-alarm rate 0.52, balanced accuracy 0.62. Reported figures for other studies are from their papers; Orengo et al. (2020) report no accuracy metric, and Kozioł et al. (2025) predict when known sites are visible rather than whether a patch holds a site, so neither is compared here.

Under the same metrics, our numbers are lower than those reported elsewhere, and the gap has two sources. First, test design: Caspari & Crespo and Barragán et al. evaluate with random splits or on training-set pixels, and Section 5.1 shows that an easier split raises our own overall AUC from 0.66 to 0.78. Second, task: Vincent et al. separate looted from preserved sites that are already known, whereas we must separate a site from ordinary ground in a region the model has never seen. The metric match alone therefore does not make the numbers comparable. A fair comparison would re-run each method on the same sites under the same region-held-out protocol. We did this for a random forest on spectral and elevation features, in the spirit of Menze & Ur (2012) and Orengo et al. (2020), and for a foundation model in the spirit of Vincent et al. (2025). We did not re-run the convolutional-network methods.

7. Limitations

- **Labels.** Classes come from fixed rules applied to EAMENA's own descriptions, without expert checking, and positions carry recording uncertainty.
- **Which sites are recorded.** EAMENA records sites that someone noticed, often on imagery, so recorded sites may be easier to see than the average site. Backgrounds may also contain unrecorded sites, which would make our estimates conservative.

- **Backgrounds.** One background design was used: three random patches per site, at least 2.5 km from any recorded site. We did not test other distances (5 km, 10 km), terrain-matched backgrounds or buffers around sites, so unrecorded sites in the backgrounds, or a model that separates raised or rough terrain from flat terrain rather than archaeology, cannot be ruled out.
- **Sample size and regions.** 196 sites in 14 regions, one of which holds 58 sites; intervals are about ±0.04.
- **Elevation.** Both elevation models have a native resolution of about 30 m, so small mounds are smoothed out. FABDEM removes buildings by machine learning, and some structures may remain.
- **Seasons.** We did not separate the four seasons, compare cloud-free with cloud-affected sites (9 of 73 winter tiles are partly obscured), test settlements against burial mounds, or measure how redundant the seasonal features are with the spectral features already extracted. The seasonal result is therefore descriptive, not explained.
- **Unequal regions and folds.** The 14 regions range from 1 to 58 sites, and some folds are dominated by one site type (Appendix A). Scores are above chance in every fold (0.63–0.73) and remain so when the two largest regions are removed (0.66), but the settlement and burial-mound results rest largely on different regions.
- **Statistical dependence between sites.** Sites in a region are not independent, and only 14 regions exist. A region-level bootstrap (Appendix B) leaves the elevation result convincing but widens the colour-only intervals to include chance.
- **Imagery.** One year (2024) of median composites; no radar and no high-resolution imagery. Chiricallo et al. (2026) show that Sentinel-1 radar adds to elevation for tell segmentation, so our elevation-based results may be a lower bound.
- **Guarantee.** The conformal guarantee assumes regions are exchangeable; it held empirically here but is not assured for very different landscapes.
- **Scope.** The zero-shot prototypes and symbolic rules of the original ArcheoZS design were not supported by the earlier 18-site study (accuracy at or below the majority baseline) and are not part of the pipeline evaluated here.

8. Future Work

- Add several hundred sites to test whether the foundation model overtakes simple features, as its learning curve suggests.
- Test the background design: distances of 2.5, 5 and 10 km, terrain-matched against random backgrounds, buffers around known sites, and natural mounds or dunes as negatives, to show whether the model detects archaeology or only unusual terrain.
- Explain the elevation result: random-forest and permutation importance computed on held-out regions only, SHAP attribution, feature ablations, and each elevation feature alone against the full set of 15, with no test-region information used.
- Explain the seasonal result: single-season ablations, cloud-free against cloud-affected sites, settlements against burial mounds, and a test of redundancy with the spectral features.
- Use finer elevation (e.g. stereo-derived or national 5–12 m models) and combine elevation with imagery in ways that do not dilute it, such as separate models or feature selection inside each fold.

- Add radar (Sentinel-1), which Chiricallo et al. (2026) found complements elevation, and declassified 1960s CORONA imagery, which shows many sites before modern disturbance.
- Run the best model across whole images and have archaeologists check the top-ranked unrecorded locations, without publishing their coordinates.

9. Conclusion

Under strict testing (same-image backgrounds, 300 km regions held out, five pooled background draws and corrected paired tests), a geospatial foundation model does not beat simple spectral statistics for finding archaeological sites, and four seasons of imagery help only modestly. Free bare-earth elevation does help, raising detection from 0.60 to 0.69 AUC for settlements and burial mounds alike. That signal comes from mound-sized relief and is not caused by modern buildings. With calibrated decisions and conformal triage, the model lets survey teams skip about a third of the sampled background patches while keeping 90% of sites. The broader lesson is about evaluation: an easier test setup or a single lucky run would have produced better-looking numbers, and a different story.

Data, Code and Ethics

Code for every experiment (data download, patch building with fingerprints, all models, pooled statistics, triage and figures) is available from the author. Sentinel-2 imagery and the Copernicus elevation model are open; FABDEM is used under its CC BY-NC-SA 4.0 licence. EAMENA records were accessed through the EAMENA database under its terms of use. Because site locations can help looters, exact coordinates are not published here, and any future list of candidate sites will be shared only with heritage authorities.

Appendix A. Fold composition and regional balance

The 14 regions are very unequal (1 to 58 sites), so the five folds are too. Table A1 gives each fold's contents and Figure A1 shows where the folds lie. Fold numbers are those of the cross-validation split, which assigns regions to folds by size. Every site appears in exactly one test fold. Backgrounds are three per site per draw, so each fold holds three times its number of sites per draw and fifteen times over the five pooled draws.

Table A1. Composition of the five region-held-out folds and the score of the headline model (random forest, 1 date + bare-earth elevation) on each.

Fold	Regions	Test sites	Settle-ments	Burial mounds	Backgrounds per draw (pooled)	Within-site AUC
1	1 (R6)	58	32	26	174 (870)	0.733
2	1 (R10)	36	1	35	108 (540)	0.722
3	2 (R1, R14)	34	33	1	102 (510)	0.663
4	5 (R2, R4, R5, R12, R13)	34	19	15	102 (510)	0.675
5	5 (R3, R7, R8, R9, R11)	34	15	19	102 (510)	0.627
All	14	**196**	**100**	**96**	**588 (2,940)**	**0.6905**

Within-site AUC is the mean over the sites of the fold, each averaged over the five background draws. The all-sites value is the Table 4 result (0.6905; shown as 0.690 there). It is the mean over all 196 sites, not the mean of the five fold values.

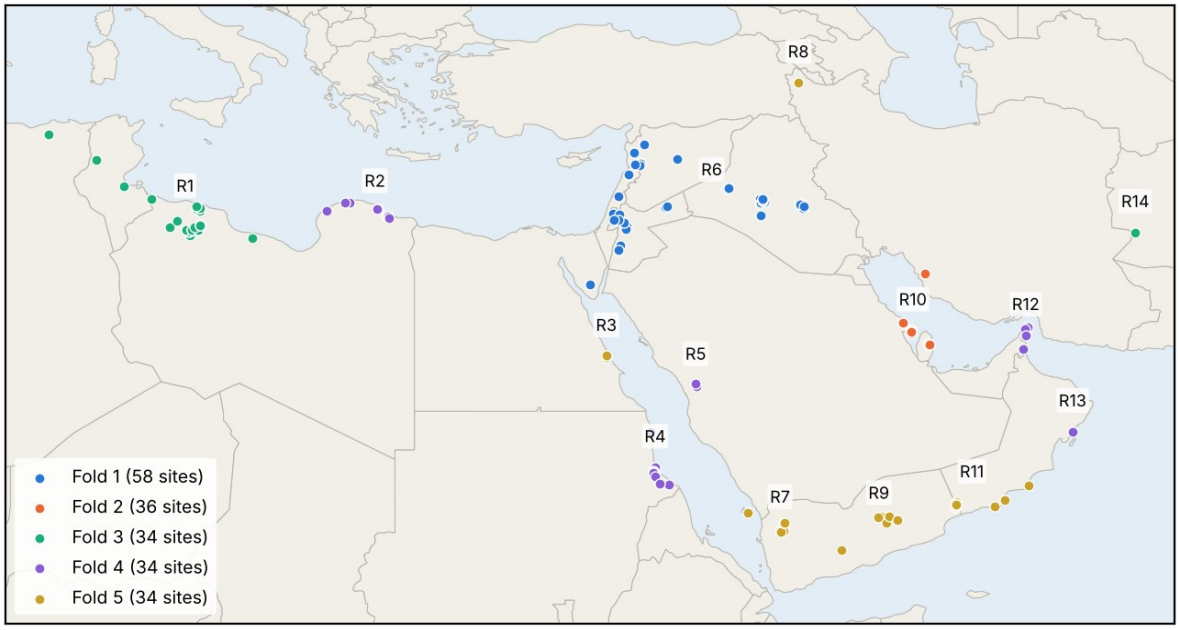


Figure A1. Test fold of every site. Colours mark the five folds, labels R1–R14 the regions (the same labels as Figure 1). Borders: Natural Earth.

Table A2. The 14 regions: size, type mix, location and score.

Region	Fold	Sites	Settle-ments	Burial mounds	Main countries (sites)	Within-site AUC
R1	3	33	32	1	Libya 29, Tunisia 3, Algeria 1	0.655
R2	4	15	14	1	Libya 15	0.662
R3	5	2	2	0	Egypt 2	0.700
R4	4	12	1	11	Sudan 12	0.672
R5	4	2	0	2	Saudi Arabia 2	0.333
R6	1	58	32	26	Iraq 32, Syria 10, Jordan 9, Palestine 4, Lebanon 2, Egypt 1	0.733
R7	5	15	10	5	Yemen 14, Saudi Arabia 1	0.671
R8	5	1	1	0	Iran 1	0.733
R9	5	6	0	6	Yemen 6	0.356
R10	2	36	1	35	Bahrain 28, Qatar 4, Saudi Arabia 3, Iran 1	0.722
R11	5	10	2	8	Yemen 6, Oman 4	0.700
R12	4	4	3	1	Oman 3, United Arab Emirates 1	0.883
R13	4	1	1	0	Oman 1	0.733
R14	3	1	1	0	Afghanistan 1	0.933

Regions are numbered from west to east. Scores for regions with fewer than 10 sites (R3, R5, R8, R9, R12–R14) are too noisy to interpret on their own.

A.1 What this shows

- **The folds are unbalanced in size and in type.** Fold 1 is the single largest region (R6, 58 sites, Iraq and the Levant). Fold 2 is the Bahrain burial-mound fields (R10; 35 of 36 sites are burial mounds). Fold 3 is almost all settlements (33 of 34, mostly Libya). Folds 4 and 5 are mixed. Our results therefore average over folds that test different things, and the burial-mound and settlement scores come largely from different regions.
- **The score is above chance in every fold,** but varies: 0.73, 0.72, 0.66, 0.68 and 0.63. The weakest is fold 5 (Yemen, Oman, Egypt and Iran); the two folds built from the largest regions score highest.
- **No single region produces the result.** Removing the largest region (R6) lowers the overall within-site AUC from 0.69 to 0.67, removing the Bahrain fold's region (R10) to 0.68, and removing both to 0.66 (102 sites). The signal weakens but remains clearly above 0.50.
- **Uncertainty is slightly wider when regions, not sites, are resampled.** A bootstrap over the 14 regions gives 0.64–0.72, against 0.65–0.73 when sites are resampled.
- **Site type is confounded with region.** Because R10 (Bahrain) holds 35 of the 96 burial mounds and R1 (Libya and neighbours) holds 32 of the 100 settlements, the 'settlements vs burial mounds' comparison cannot be separated cleanly from regional differences.

Appendix B. Region-level bootstrap

The main intervals resample individual sites. Sites in the same region share terrain, imagery and history, so they are not independent, and site-level intervals may be too narrow. After review, we repeated every interval and every paired test in Tables 4 and 5 with two resampling schemes that respect the region structure, using the same pooled per-site scores and 5,000 resamples each. This second analysis was added after the main results were known; it is a robustness check, not a pre-planned test.

- **Region (cluster) bootstrap:** draw 14 regions with replacement and keep all their sites. This treats each region as one unit.
- **Two-stage (hierarchical) bootstrap:** draw 14 regions with replacement, then resample sites with replacement within each drawn region. This adds within-region variation on top of the between-region variation, and tends to give the widest, most cautious intervals.

Differences are always paired: the same resampled sites are used for both models. P values are one-sided and corrected with the Holm method within each family, as in Table 5.

Table B1. Within-site AUC with three 95% intervals (196 sites, five draws pooled).

Method	AUC	Site bootstrap	Region bootstrap	Two-stage bootstrap
Random forest, 1 date	0.599	0.56–0.64	0.51–0.65	0.50–0.66
Random forest, 4 seasons	0.633	0.59–0.67	0.56–0.68	0.55–0.69
Prithvi (frozen), 1 date	0.579	0.54–0.62	0.51–0.65	0.50–0.66
Prithvi (frozen), 4 seasons	0.611	0.57–0.65	0.54–0.67	0.53–0.67
Prithvi (frozen), 4 seasons, per season	0.618	0.58–0.65	0.55–0.66	0.54–0.67
Elevation only (FABDEM)	0.680	0.64–0.72	0.64–0.70	0.63–0.72
Random forest, 1 date + elevation	**0.690**	0.65–0.73	0.63–0.72	0.63–0.74
Random forest, 4 seasons + elevation	0.680	0.64–0.72	0.60–0.73	0.60–0.74

Table B2. Paired comparisons under the three resampling schemes: 95% interval and Holm-corrected P.

Comparison	Diff.	Site bootstrap	Region bootstrap	Two-stage bootstrap
Prithvi: 4 seasons minus 1 date	+0.032	−0.006 to +0.069; P 0.152	−0.006 to +0.075; P 0.128	−0.021 to +0.092; P 0.327
Random forest: 4 seasons minus 1 date	+0.037	+0.003 to +0.071; P 0.067	+0.013 to +0.075; P 0.002	−0.004 to +0.092; P 0.162
4 seasons: Prithvi minus random forest	−0.022	−0.067 to +0.023; P 1.000	−0.083 to +0.056; P 1.000	−0.097 to +0.069; P 1.000
4 seasons, per season: Prithvi minus random forest	−0.015	−0.057 to +0.028; P 1.000	−0.071 to +0.060; P 1.000	−0.085 to +0.071; P 1.000
1 date: + elevation minus without	**+0.091**	+0.059 to +0.125; P 0.001	+0.064 to +0.138; P 0.001	+0.049 to +0.152; P 0.001
4 seasons: + elevation minus without	**+0.047**	+0.024 to +0.069; P 0.001	+0.024 to +0.066; P 0.001	+0.012 to +0.076; P 0.006
Elevation only minus chance (0.50)	**+0.180**	+0.142 to +0.216; P 0.001	+0.141 to +0.203; P 0.001	+0.126 to +0.222; P 0.001

P is the Holm-corrected one-sided P. With 5,000 resamples the smallest uncorrected P is 0.0002, so a value shown as 0.001 is at or near the smallest the method can show.

B.1 What changes

- **The elevation conclusion holds under all three schemes.** The gain from adding elevation to one date of imagery is +0.091, with intervals of +0.059 to +0.125 (sites), +0.064 to +0.138 (regions) and +0.049 to +0.152 (two-stage). Holm P stays at 0.001. Elevation alone stays above chance by 0.13 to 0.22 in every scheme. The gain with four seasons is the least certain of the three (+0.012 to +0.076 at the two-stage level, P 0.006) but remains convincing.
- **The elevation gain does not depend on one region.** Leaving out each of the 14 regions in turn gives a gain between +0.082 and +0.106. The gain is positive in 11 of the 14 regions; the three where it is negative hold 2, 2 and 4 sites.
- **Colour-only intervals widen a lot.** For the one-date random forest the interval grows from 0.56–0.64 to 0.51–0.65 (regions) and 0.50–0.66 (two-stage), and the lower end reaches chance. The same holds for frozen Prithvi. The finding that colour alone is a weak signal is therefore less certain than the site-level interval suggested, in both directions: it may be a little useful or not useful at all.
- **Elevation intervals widen far less** (0.65–0.73 to 0.63–0.72 for regions), because the elevation signal is similar across regions.
- **The seasonal gain is not stable across schemes.** For the random forest, four seasons minus one date is +0.037 in all three, but is convincing under the region bootstrap (Holm P 0.002), not under the site bootstrap (0.067) or the two-stage bootstrap (0.162). We therefore keep describing it as a likely but unconfirmed improvement. For Prithvi it is not convincing under any scheme.
- **The comparison between the foundation model and the random forest is unchanged:** no difference is detectable under any scheme, with intervals that cover zero.

B.2 Limitations of this analysis

- **Fourteen regions is few.** Percentile bootstrap intervals based on 14 clusters tend to be too narrow, so even the region-level intervals may understate the true uncertainty.
- **The regions are very unequal** (1 to 58 sites; Appendix A). A region with one site contributes almost nothing on its own, and the largest regions drive the result in many resamples. Many resamples contain the 58-site region more than once, or not at all, which widens the intervals.
- **The two-stage bootstrap is deliberately cautious.** Resampling within regions as well as between them counts some variation twice, so its intervals are wider than needed. We read the region and two-stage columns as a range, not as competing answers.
- **The same folds are used throughout.** The bootstrap does not re-train models, so it captures variation in which sites and regions are sampled but not variation from other fold assignments.

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